

Ex. 3.37 | Ch 3 | P108 | Chloé/2025.7.28

证明: 位力定理 $\frac{d}{dt}\langle xP \rangle = 2\langle T \rangle - \langle x \frac{\partial V}{\partial x} \rangle$

$$\frac{d}{dt}\langle xP \rangle = \frac{1}{i\hbar} \langle [xP, H] \rangle + 0 = \frac{1}{i\hbar} \langle [x, \frac{P^2}{2m} + V(x)] \rangle = \frac{1}{i\hbar} \langle [xP, \frac{P^2}{2m}] + [xP, V(x)] \rangle$$

$$\text{先计算 } [xP, \frac{P^2}{2m}] = x[P, \frac{P^2}{2m}] + [x, \frac{P^2}{2m}]P = x \cdot 0 + \frac{1}{2m} [x, P^2]P = \frac{1}{2m} \cdot 2i\hbar P \cdot P = i\hbar \frac{P^2}{m}$$

$$\text{再计算 } [xP, V(x)] = x[P, V(x)] + [x, V(x)]P = x \cdot (-i\hbar \frac{\partial V}{\partial x}) + 0 = -i\hbar x \frac{\partial V}{\partial x}$$

$$\text{故 } \frac{d}{dt}\langle xP \rangle = \frac{1}{i\hbar} \langle i\hbar \frac{P^2}{m} + (-i\hbar x \frac{\partial V}{\partial x}) \rangle = \langle \frac{P^2}{m} \rangle - \langle x \frac{\partial V}{\partial x} \rangle$$

$$= 2\langle T \rangle - \langle x \frac{\partial V}{\partial x} \rangle \quad \text{得证}$$

2. 对定态 $\psi(x, t) = \psi(x)e^{-iEt/\hbar}$, $\langle xP \rangle$ 不显含时间, 故 $\frac{d}{dt}\langle xP \rangle = 0$ 3. 证明: 谐振子定态有 $\langle T \rangle = \langle V \rangle$

$$V(x) = \frac{1}{2}m\omega^2 x^2, \quad \frac{\partial V}{\partial x} = m\omega^2 x$$

$$\text{由 } 2\langle T \rangle = \langle x \frac{\partial V}{\partial x} \rangle = \langle x \cdot m\omega^2 x \rangle = \langle m\omega^2 x^2 \rangle = m\omega^2 \langle x^2 \rangle = 2\langle V \rangle$$

$$\text{故 } \langle T \rangle = \langle V \rangle$$

习题 2.11 中, $\langle T \rangle_0 = \langle V \rangle_0 = \frac{1}{4}\hbar\omega$, $\langle T \rangle_1 = \langle V \rangle_1 = \frac{3}{4}\hbar\omega$, 题干是谐振子,

与位力定理一致。